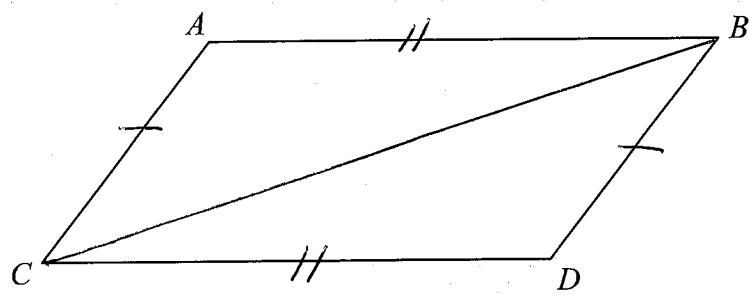
1 *ABCD* is a parallelogram



Prove that triangle *ABC* is congruent to triangle *BCD*.

AC = BD opposite sides of a parallelogram are equal

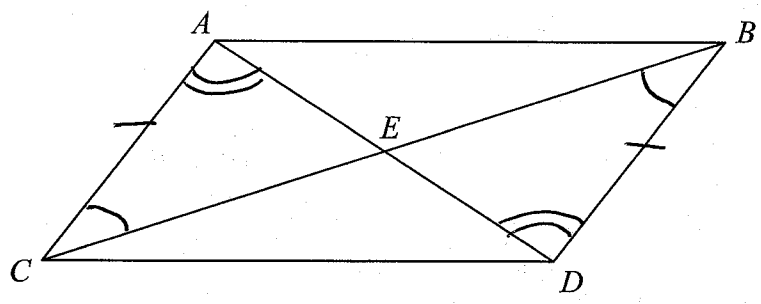
AB = CD ——— " ———

BC is common to both triangles

SSS

(Total for Question 1 is 3 marks)

2 *ABCD* is a parallelogram
E is the point where the diagonals *AD* and *BC* meet.



Prove that triangle *ACE* is congruent to triangle *BDE*.

AC = BD opposite sides of a parallelogram are equal

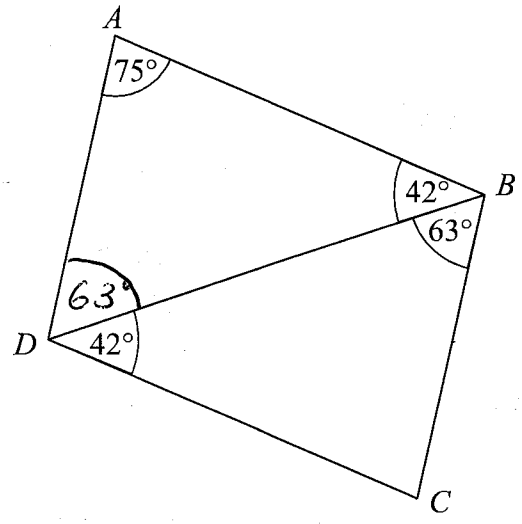
ACB = CBD Alternate angles are equal

CAD = ADB ——— " ———

ASA

(Total for Question 2 is 3 marks)

3 The diagram shows two triangles, *ABD* and *BCD*.



Prove that triangle *ABD* is congruent to triangle *BCD*.

ADB = 180 − 75 − 42

= 63°

Angles in a triangle add to 180°

BD is common to both triangles

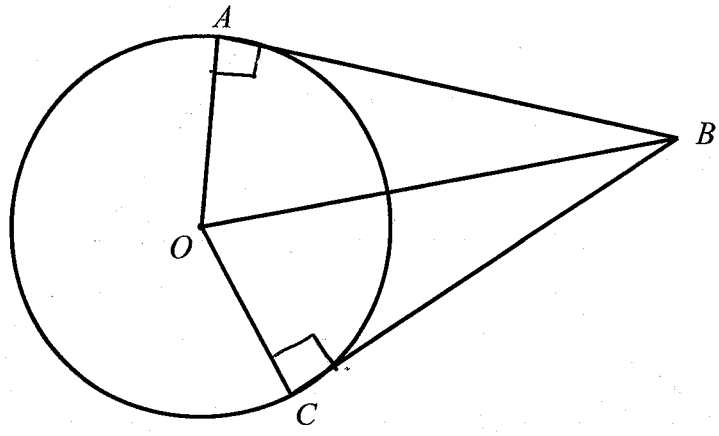
ABD = BDC

CBD = ADB

ASA

(Total for Question 3 is 3 marks)

4 A and C are points on a circle, centre O.
AB and BC are tangents to the circle.



Prove that triangle ABO is congruent to triangle BCO.

OCB = OAB = 90° Tangent meets radius at 90°

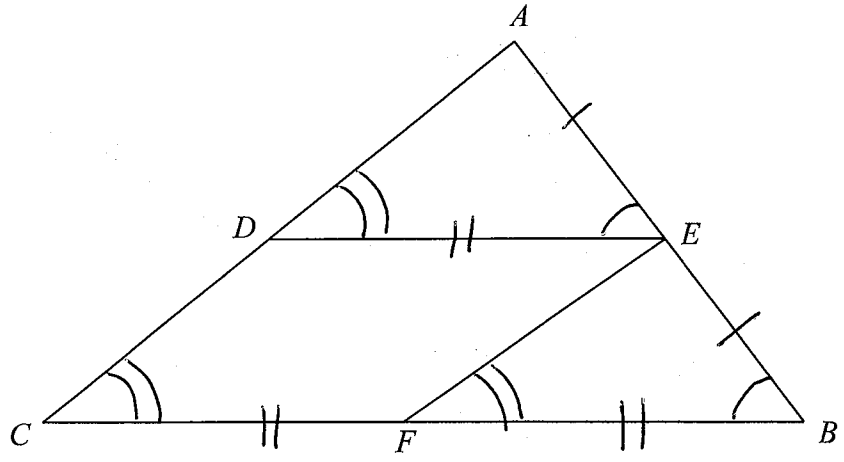
OB is common to both triangles

OA = OC both a radius

RHS

(Total for Question 4 is 4 marks)

5 *ABC* is a triangle.



CDEF is a parallelogram such that:
D is the midpoint of *AC*
E is the midpoint of *AB*
F is the midpoint of *BC*

Prove that triangle *ADE* is congruent to triangle *BEF*.

AE = BE ~~A~~ E is the midpoint of AB

AED = EBF corresponding angles are equal

CF = DE opposite sides of a parallelogram are equal

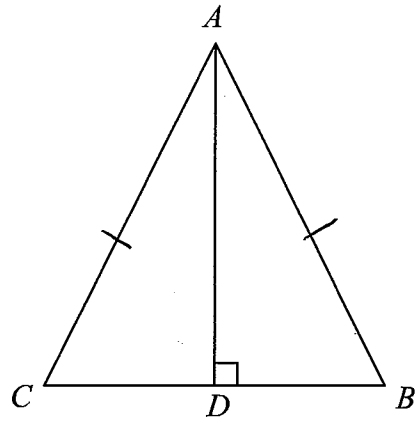
CF = BF F is the midpoint of BC

∴ DE = BF

SAS

(Total for Question 5 is 4 marks)

6 *ABC* is an equilateral triangle.



D lies on *BC*
AD is perpendicular to *BC*

Prove that angle *CAD* is equal to angle *BAD*.

AB = AC lengths of equilateral triangle are equal

AD is common to both triangles

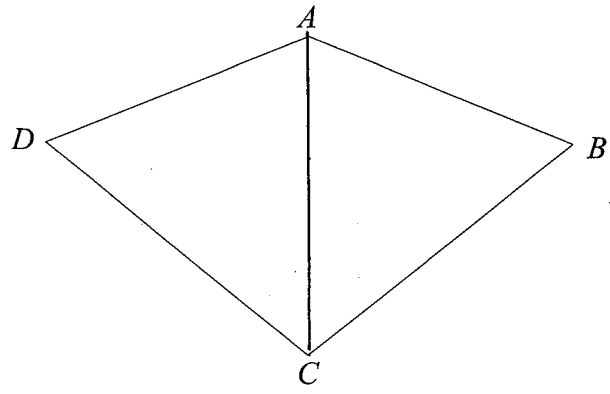
Triangle ABD and triangle ACD are right angled (given)

Triangle ABD and triangle ACD are congruent RHS

∴ Angle CAD = Angle BAD

(Total for Question 6 is 4 marks)

7 *ABCD* is a quadrilateral



$AB = AD$
$BC = CD$

Prove that angle *ABC* is equal to angle *ADC*.

Triangle ABC and triangle ACD are congruent

AB = AD given

BC = CD given

AC is common to both triangles

SSS

∴ Angle ABC = Angle ADC

(Total for Question 7 is 4 marks)